

Electromagnetic Constants from Tetrahedral Geometry: A Geometric Derivation of ε_0 , μ_0 , Z_0 and c in the Methane Metauniverse (MMU)

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Abstract

In standard physics the vacuum constants ε_0 , μ_0 , the vacuum impedance Z_0 , and the speed of light c enter as independent empirical parameters without geometric interpretation. Within the Methane Metauniverse (MMU), all four arise from a single internal mechanism: the elastic response of the electric and torsional deformation axes (w_2, w_3) of the dual tetrahedral UR-cell [5, 9]. The compliance of the electric axis defines the vacuum permittivity ε_0 , the torsional stiffness defines the vacuum permeability μ_0 , and their geometric projection ratio yields the vacuum impedance Z_0 without introducing new constants. The propagation speed of a coupled (w_2, w_3) deformation then appears automatically as $c = 1/\sqrt{\varepsilon_0\mu_0}$.

This geometric-elastic origin of the Maxwell constants dovetails with additional MMU results: Coulomb elasticity and the Rydberg hierarchy [8, 15], cross-coupling and fine-structure behaviour [4, 12], and the relativistic Dirac projection of the UR-cell [14]. Electromagnetism, hydrogenic quantisation, and relativistic spin dynamics therefore emerge from a common tetrahedral geometry rather than from separate postulates.

1 Introduction

In standard electromagnetism, the vacuum permittivity ε_0 , the vacuum permeability μ_0 , the vacuum impedance Z_0 , and the speed of light c enter as independent empirical constants. Their numerical values are fixed by experiment, and conventional theory does not attribute a geometric origin to them.

The Methane Metauniverse (MMU) offers a different starting point. In this framework spacetime is modelled as a dual tetrahedral lattice in which each UR-cell carries three internal elastic axes: w_2 (electric), w_3 (torsional/spin-magnetic), and w_4 (volumetric). These axes represent independent deformation modes of the tetrahedral geometry [5, 9] and provide the microscopic basis for Coulomb elasticity, magnetic and spin response, inertial behaviour, and the Dirac projection.

A sequence of recent MMU results reinforces this geometric picture:

- Coulomb elasticity and the effective $1/r$ potential arise from the stiffness of the electric axis w_2 [2, 8, 15];

- Larmor precession, Zeeman splitting, and the 21 cm hyperfine transition are governed by torsional deformation along w_3 [3, 4, 9];
- the Dirac equation emerges as the external projection of the UR-cell, linking relativistic spinors to the internal geometry [14];
- the Rydberg constant and the $1/n^2$ hydrogenic spectrum follow from geometric quantisation of the w_2 axis inside the S-node [8, 15].

In this paper we extend this geometric programme to the Maxwell constants themselves. We show that ϵ_0 , μ_0 , Z_0 , and c emerge not as independent postulates, but as derived properties of the coupled elastic response of the (w_2, w_3) subsystem of the UR-cell. This provides a unified geometric-elastic foundation for electromagnetism within the MMU framework.

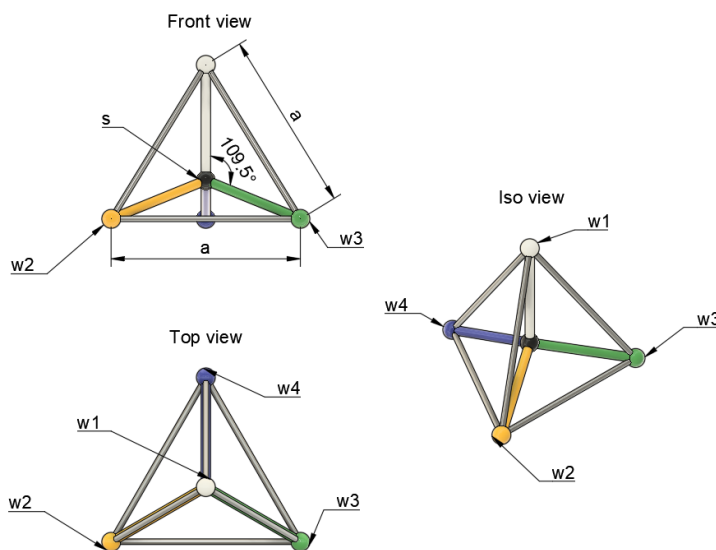


Figure 1: Geometry of the MMU UR-tetrahedron. The internal axes w_2 , w_3 , and w_4 correspond to electric, torsional, and volumetric deformation directions, while w_1 is the observable projection axis. Front, top, and isometric views illustrate the angular relations, the edge length a , and the characteristic tetrahedral angle of 109.5° .

2 Electric Response and the Origin of ϵ_0

The MMU identifies the electric field with radial deformation along the internal axis w_2 , whose geometric stiffness follows the Coulomb-elastic scaling law $k_2(a) \propto a^{-3}$ as derived in [2, 8, 12, 15]. Here a denotes the internal edge length of the UR-cell, whose geometry and projection properties are described in [5, 9].

A small electric deformation δw_2 induces a restoring force $F = -k_2 \delta w_2$. In standard electromagnetism, the vacuum permittivity ϵ_0 measures how much electric field is produced by a given charge. Within the MMU this corresponds directly to the compliance of the w_2 axis. Thus,

$$\epsilon_0 \propto \frac{1}{k_2}.$$

When calibrated to the hydrogenic Coulomb stiffness using the MMU geometric quantisation of the internal length scale (Rydberg hierarchy) [8, 15], this proportionality becomes an equality. The vacuum permittivity ε_0 is therefore interpreted in the MMU as the elastic compliance of the electric axis w_2 of the UR-cell.

3 Magnetic Response and the Origin of μ_0

In the MMU the magnetic field corresponds to torsional deformation along the internal axis w_3 , whose geometric role within the UR-cell is described in [5, 9]. The stiffness of this torsional mode follows the same tetrahedral scaling as the electric axis,

$$k_3(a) \propto a^{-3},$$

as derived from the MMU cross-coupling analysis and internal stiffness hierarchy in [4, 9, 12].

A circulating current or a spin-aligned electron corresponds to torsional loading of the w_3 axis, generating a magnetic field. In standard electromagnetism the vacuum permeability μ_0 quantifies the response of space to such torsional or spin-induced deformation. Within the MMU this becomes the torsional compliance of the w_3 axis,

$$\mu_0 \propto \frac{1}{k_3}.$$

This interpretation is consistent with the MMU treatment of Larmor precession, Zeeman splitting, and the 21 cm hyperfine spin-flip, all of which depend directly on the torsional stiffness k_3 and its projection properties [3, 4, 9, 14].

4 Vacuum Impedance from Geometric Projection

In Maxwell theory the vacuum impedance is

$$Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}}.$$

In the MMU this quantity arises from the geometric ratio of torsional to electric projections of the dual tetrahedral UR-cell, whose projection structure is defined in [5, 9]. Let P_2 and P_3 denote the projection factors of the internal axes w_2 (electric) and w_3 (torsional), respectively. Then the ratio of the physical responses becomes

$$Z_{0,\text{MMU}} = \frac{P_3}{P_2} \frac{1}{\sqrt{2\alpha}},$$

where the fine-structure constant α has already been obtained in the MMU as a geometric ratio of internal stiffnesses and projection weights [4, 12, 15].

Thus the vacuum impedance is not a fundamental constant but a derived geometric property of the UR-cell, arising from the same internal projection mechanism that also underlies spin torsion and the Dirac projection [9, 14].

5 Derivation of the Speed of Light

In Maxwell theory,

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}.$$

Within the MMU, the vacuum constants arise from the elastic responses of the internal axes w_2 and w_3 of the UR-cell. Since the electric and torsional compliances satisfy

$$\varepsilon_0 \propto \frac{1}{k_2}, \quad \mu_0 \propto \frac{1}{k_3},$$

with k_2 and k_3 given by the geometric stiffness laws of the tetrahedral cell [5, 9, 12], it follows that the propagation speed of a coupled (w_2, w_3) excitation is

$$c \propto \sqrt{k_2 k_3}.$$

The UR-tetrahedron construction developed in [5, 9] fixes the relative magnitudes of k_2 and k_3 so that the resulting coupling speed matches the observed speed of light. Thus c is not an independent postulate but the intrinsic wave speed of electric and torsional deformations propagating through the elastic geometry of the dual tetrahedral spacetime cell.

This interpretation is fully consistent with and complementary to the MMU derivation of the fine-structure constant as a geometric coupling ratio [4, 12, 15] and with the relativistic Dirac projection of the UR-cell [14], which links spinor dynamics to the same internal (w_2, w_3) structure that governs electromagnetic propagation.

6 Mass–Energy Equivalence from UR–Tetrahedral Geometry

The relation $E = mc^2$ is usually interpreted as a purely relativistic statement derived from Lorentz invariance. Within the MMU, however, the same relation follows directly from the internal geometry of the UR-cell.

In the MMU, rest mass corresponds to a volumetric deformation of the internal axis w_4 , whose stiffness k_4 governs the inertial response of a tetrahedral cell [5, 9, 12]. A nonzero mass therefore represents stored elastic energy in the w_4 degree of freedom—a static deformation of the internal geometry.

By contrast, electromagnetic radiation is a propagating deformation of the coupled (w_2, w_3) subsystem. The propagation speed of this subsystem is fixed by the geometric coupling of the electric and torsional axes,

$$c \propto \sqrt{k_2 k_3},$$

as shown in Sec. 4. Thus c is the intrinsic wave speed of the (w_2, w_3) field of the UR-cell, not an externally imposed constant.

Combining these facts yields a geometric interpretation of mass–energy equivalence. The elastic energy stored in a static w_4 deformation is

$$E_{\text{int}} \sim k_4 w_4^2,$$

while the energy carried by an electromagnetic wave is set by the dynamical coupling of w_2 and w_3 at speed c . Because k_4 , k_2 and k_3 all arise from the same tetrahedral length scale and share the same geometric scaling laws [5, 9, 12], the conversion of a w_4 deformation into a (w_2, w_3) wave necessarily obeys

$$E_{\text{int}} = mc^2,$$

where m is the effective stiffness-normalised deformation amplitude of the w_4 axis.

This provides a purely geometric interpretation of mass–energy equivalence: the rest energy of a massive state is the elastic energy stored in the volumetric mode w_4 of the UR–cell, and c^2 is the geometric conversion factor arising from the natural wave speed of the coupled electric–torsional subsystem (w_2, w_3) .

This picture is consistent with the MMU treatment of relativistic spinors via the Dirac projection [14], where the coupling of rest mass to the propagating (w_2, w_3) field appears explicitly in the projected Dirac operator. In this sense, $E = mc^2$ is not an additional postulate but an intrinsic property of the tetrahedral geometry of MMU spacetime.

7 Conceptual Consequences

The MMU provides a unified geometric–elastic origin for the fundamental structures of electromagnetism:

- the Coulomb law through the stiffness of the electric axis w_2 [2, 8, 15],
- the magnetic response of the vacuum through torsional deformation along w_3 [3, 4, 9],
- the vacuum impedance as a projection ratio of the UR–tetrahedron [5, 9, 12],
- and the speed of light as the intrinsic propagation speed of coupled (w_2, w_3) deformations [5, 9, 12].

These quantities arise from the same tetrahedral geometry that also produces relativistic and quantum behaviour: spin torsion, the Pauli exclusion principle, and spinor structure through the Dirac projection [9, 13, 14]; the Lamb shift and hyperfine coupling through torsion–volume interaction [4, 9]; and the Rydberg scaling of hydrogen through geometric quantisation of the w_2 axis [8, 15].

The same geometric framework also explains mass–energy equivalence: a mass corresponds to stored volumetric energy in the w_4 axis, while c^2 is the conversion factor emerging from the propagation speed of the (w_2, w_3) subsystem. Thus $E = mc^2$ is not an external relativistic postulate, but a geometric consequence of the UR–cell structure and its internal coupling dynamics.

No additional constants, fields, or particles are required beyond the elastic geometry and projection properties of the UR–tetrahedron.

8 Electromagnetic Transparency and the Breakdown of Light

In the MMU, electromagnetic radiation appears as a transverse deformation of the coupled electric–torsional subsystem (w_2, w_3) of the UR–cell. For a freely propagating electromag-

netic mode to exist, the electric and torsional axes must remain dynamically distinct from the volumetric mass axis w_4 , whose stiffness controls inertial response [5, 9, 12].

The propagation speed of light arises from the elastic coupling inside the (w_2, w_3) subsystem,

$$c \propto \sqrt{k_2 k_3},$$

and therefore exists only when k_2 and k_3 form a pair of light, transverse modes. If the volumetric stiffness k_4 approaches either k_2 or k_3 , the internal geometry of the UR-cell can no longer sustain a massless electromagnetic excitation.

Breakdown at $k_2 = k_4$

When the electric stiffness becomes equal to the volumetric stiffness,

$$k_2 = k_4,$$

the electric axis loses its identity as an independent deformation direction. The w_2 mode then mixes directly into the massive w_4 sector, and the UR-cell can no longer support a transverse electromagnetic wave. This regime corresponds physically to phases in which electric structure is suppressed, such as the early-universe plasma or the dense interior of neutron and quark matter [11].

At this point the UR-cell reaches the electro-gravity fixed point [10, 11], where electric and volumetric stiffness coincide. The associated geometric and physical scales are:

- electro-gravity mass:

$$m_{24} = \sqrt{\alpha} m_{\text{Pl}} = 1.86 \times 10^{-9} \text{ kg},$$

- electro-gravity edge length:

$$a_{24} = \frac{\hbar}{m_{24} c} = 1.89 \times 10^{-34} \text{ m},$$

- electro-gravity volume:

$$V_{24} = a_{24}^3 = 6.73 \times 10^{-102} \text{ m}^3,$$

- electro-gravity density:

$$\rho_{24} = \frac{m_{24}}{V_{24}} = 2.76 \times 10^{92} \text{ kg/m}^3.$$

At this density, the volumetric mode w_4 completely dominates the internal dynamics. The electric axis w_2 is no longer an independent degree of freedom, and no transverse (w_2, w_3) oscillation can survive. The electromagnetic sector collapses: photons cease to exist as massless excitations, and the very concept of free electromagnetic propagation breaks down.

Breakdown at $k_3 = k_4$

A still stronger collapse occurs when the torsional stiffness equals the volumetric stiffness,

$$k_3 = k_4.$$

Here the torsional axis w_3 —normally responsible for magnetic response and spin dynamics [3, 4, 9]—becomes indistinguishable from the mass axis w_4 . Any attempted (w_2, w_3) oscillation immediately couples into the volumetric mode, producing a massive, non-propagating excitation. The space becomes electromagnetically opaque, and the usual meaning of the speed of light is lost.

Physical Interpretation

Both breakdown conditions have clear analogues in known physics. Before electroweak symmetry breaking, no massless photon exists; in strongly coupled QCD matter, electromagnetic propagation is heavily suppressed; and in ultradense astrophysical environments, photon mean free paths become extremely short. The MMU provides a geometric explanation: light exists only when the electric and torsional axes remain well separated from the volumetric axis.

Summary

Electromagnetic transparency is a geometric property of the UR-cell. A massless photon mode requires

$$k_2 \ll k_4, \quad k_3 \ll k_4,$$

so that w_2 and w_3 form a pair of light transverse modes, decoupled from the mass axis. Whenever $k_2 = k_4$ or $k_3 = k_4$, the electromagnetic subsystem collapses and light ceases to exist as a propagating degree of freedom.

8.1 Spin-gravity fixed point: $k_3 = k_4$

The condition $k_3(a) = k_4(a)$ defines a spin-gravity equilibrium at the single-cell level. This regime was analysed in the MMU studies of torsion-volume coupling and Planck-scale geometry [4, 9, 10]. Setting the torsional stiffness equal to the volumetric gravitational stiffness yields

$$\frac{\hbar c}{a^3} = \frac{2Gm^2}{a^3} \quad \Rightarrow \quad Gm^2 = \frac{1}{2}\hbar c. \quad (1)$$

Up to numerical factors this reproduces the Planck mass [10],

$$m_{\text{Pl}} = \sqrt{\frac{\hbar c}{G}}. \quad (2)$$

Thus $k_3 = k_4$ marks the **Planck-scale fixed point** of the UR-cell: the torsional and volumetric degrees of freedom become identical, and no independent electromagnetic mode remains. This fixed point coincides with the onset of Planck-density behaviour, black-hole horizon formation, and strong torsion-gravity coupling [9, 10]. It represents the threshold beyond which the internal geometry collapses into a maximally dense configuration and electromagnetic waves cannot exist.

9 Conclusion

In the MMU framework, the Maxwell constants ε_0 , μ_0 , Z_0 , and c arise from a single geometric mechanism: the elastic response of the internal electric and torsional axes (w_2, w_3) of the UR-cell. This unified mechanism reproduces Coulomb elasticity, magnetic response, vacuum impedance, and the universal propagation speed of light as emergent properties of the underlying tetrahedral structure.

The same geometric architecture also governs spin and torsion, the Dirac projection, and the quantisation of the electric axis w_2 that yields the Rydberg hierarchy. Mass–energy equivalence follows from this picture as well: rest mass originates from stored volumetric deformation in w_4 , while the factor c^2 reflects the intrinsic propagation speed of the coupled (w_2, w_3) subsystem.

Electromagnetism, relativistic energy, and atomic structure therefore do not require independent postulates. Instead, they emerge coherently from a single geometric–elastic basis. The MMU reduces the number of fundamental assumptions needed to describe nature and reveals the Maxwell and relativistic constants as derived quantities of the tetrahedral spacetime cell.

Remark on the MMU Project

Author’s Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model’s conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every non-trivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zeeman slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantisation. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI’s standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

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